BIF401 Current Paper MCQS and Questions

1. Genomics , evolutionary studies and system biology are application of bioinformatics

2. Exact matching require which type of nucleotide-same number of nucleotide

3. Accurate solution of protein structure is one of the ___ toughest problems

4.Tract back is overlap matches can start from any position in scoring matrix

5.The matrix has positive scores

6. Bottom Up Proteomics___can search people mass finger printing and shotgun proteomics data set

7. Spectrometer measures the protein by_____Molecular weight or Mass/charge

8. Simulation of the folding process depends_____Energy function

9. How genomes are available on gene bank

10 which protein is used to measure the molecules weight of intact proteins__MS1

11. PDB ____ coordinates of alpha carbon in the protein backbone can be used for comparison

13.in structure visualization c-alpha atoms are traced to recreate a _____3D protein structure

14 in UPGMA distance is calculated btw _____ between two clusters and between two trees

15.pk value of Aspartic acid 3.9

16.largest amino acid Tryptophan

17. II STRUCTURE of proteins can be obtained by using_____

18.NJ Abgasthim strategy is used to predict__RNA 2` structure

19. The loop of hair pin must be atleast___ in length 2 aminoacids

20. Fold recognition is also called .....threading

21. Ab ignition method in contrast ,base their prediction on physical models.....for these ,mechanism

22. Bioinformatics require.....smart mind, and connected to internet, both, none

23. Sequence alignment tool is.....

24., MS1,MS2 provide us data identifying unknown.....Proteins

26.MGF files develop as an open standard for..... proteomics data

27,.....PDB..coordinates of alpha carbon in protein

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28.amino acid back bone can be used for comparison.....

29.alpha carbon atom can be obtained from.....PDB

30.in formation of 2nd structure of protein C & N can make.....Hydrogen bonds

31.information for protein folding is into its native structure is in..... protein's amino acid sequence!

32.energy released during bond formation.

33.hydrogen bonding occur in ...secondary ..Protein structure

2. Bond between C and H in alpha helices____Hydrogen bonding

3. Amino acid having 3 codons_____ isoleucine

5. MALDI typically adds a protein to the _____Proton

6. low quality match gets_____lower score

7. ____ are sequences of peptides produced during ____

8. MS2 data can be used to extract____ peptide __sequence tags.

9. The alpha helices propensity is more then____Beta (not sure)

10. How many types are of protein sequencing__5 (notsure)

11. Blast can find sequences in _____

12. If a protein sequence of 26 aminoacids is fragmented at 11 aminoacid C11, Z25

1 .to find out unknown sequence of nucelotide we use . NGS or Mass spectrometry

2 how many types of peptide May be injected in mass spectrometry chamber.. . three hundred thousand to four hundred thousand

3 .....MASCOT.can search peptide mass finger printing and shotgun proteomics dataset.

4 mRNA does not have a.....

5.stabilizing and destabilizing energy are.... .

6.fasman algorithms is based on occurrence of ..... Amino Acids in known structures

7.DP recombinant the nucleotide recombination through process of ..... Traceback

8..MALDI.....add proton to a protein and a peptide..

9.positively charged amino acids are..... 3

0 Aromatic amino acid include ..... . phenylalanine, tyrosine tryptophan

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11 what is homology modelling meant.. Creating atomic model of target protein (notsure)

12.how many forces are involved in protein folding. 4

13.pairwise alignments tells the similarity between sequence... the sequences by maximizing the matches.

14.Dot plots employ dot matrix with two sequence plotted in it.....pairwise alignment and comparison.

Subjective

1. Diff b/w swiss port and TrEMBL data bases (2)
2. Give short example of how chou fasman alogrithim is used to predict the sec structure of protein (2)
3. Diff b/w shortgun and peptide mass finger printing (2)
4. Would you please elist few need of bioinformatics (3)
5. How can you compare biological sequences with each other? Elaborate on stratiges (3)
6. Elist plants mentioned in flow chart of bottom up protemics (3)
7. Givea summary of how chov fasman algorithm can be used to predict a region in Alpha helix (5)
8. OH group in RNA make it less stable than DNA explain it with the help of diagram (5)
9. Define following
   Folded protein is more stable than denatured protein
   Role of amino acids
   Factores involve in protein folding (5)
10. Please enlist seven steps of homology modeling (10)
11. What happened when ph is less than pk (2)
12. Which factors is consider during nussinov Jacobson trace back (2)
13. How does homology modeling helps to predict the sequence of aa (2)
14. Scope of bif (3)
15. Conditions involved in beta sheets (3)
16. Diff b/w ms and raw file formate (5)
17. Optimal energy function in protein structure prediction (5)
18. Rooted and unrooted tree and its uses (5)
19. Advantages and disadvantages of ab initio modeling (10)
20. How fragment protein used (2)
21. Explain one aminoacid having many codons
22. How protein structures are identify (5)
23. Phylogentry tree types(2)
24. Chou altherum apha helic summary (5)
25. Differnence b/w acidic and basic aminoacid
26. Hydrophobicity of amino acid
27. Write optimal energy function in protein structure prediction marks ; 05
28. How protein sequence identified? Marks ; 05
29. Write 4 steps of FASTA algorithm marks; 05
30. Write benefits of phylogenetic studies marks ; 03

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31. Write protocol of bottom down proteomics marks; 03
32. Write five types of RNA and their function marks ; 3
33. How homology modeling help to predict sequence of unknown proein sequence marks ;2
34. how loops are found after the finding of alpha and beta sheets?
35. How protein structure identified?
36. What happens when pH is < or > than pK?
37. What is ab inition modelling explain its process??
38. What is FASTA algorithms write down its four steps that who y arranged ur nucleotide sequence???
39. write 5 types of RNA and also write their function
40. how spectrometer work ??
41. write 3 condition neccery for the confermation of beta turn according chou fasman algorithms??
42. Limitations of ab initio 2 marks
43. Types nd functions of rna 3 marks
44. Prpsight pattern dia gya tha protein sequnce find out krna tha.5 marks
45. Fasman algo for alpha.5 marks.
46. Scoring scheme for protein sequence tags
47. Q. Seven steps of Homology Modeling [10]
48. Q. Enlist different protein sequence databases [3]
49. Q. Explain summary of Alpha helices by chou-fasman [5]
50. Q. Enlist three hydrophobic amino acids [3]
51. Q. Why we need to make protein structure when we have x-ray and NMR techniques[2]
52. Q. 3 characteristics for the formation of beta helecis by chu-fasman algorithm [2]
53. Q. Ionization with reference to protein [3]
54. Q. Explain Acidic and basic aminoacids, advantage of folded protein over denatured protein and roll of aminoacids
55.
56. Questions .
57. 1what is the phenomian of redency of coding and mention any one amino acid that encoded by three differnt codons.5 marks. .
58. 2how we can build scoring matrices .
59. 3 marks. 3.major steps involve in protein sequencing. 3 marks.
60. 4.describe elements PDB files. 1 title . 2source. 3revdat. 4 compnd .2 marks.....
61. advantages and disadvantages of in ab intio ...10 marks.
62. Seven steps of homology modelling [10]
63. difference between shotgun and f.p[2]
64. Protein fold, forcess and function of aminoacids
65. 3 funtions of BLAST
66. write down 5 bioinformatics organization that employe
67. three types of codon
68. how loops are formed after the bludging of Alpha and Beta sheets.
69. Public databases that commonly used for proteins

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